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Z → τ_hτ_h CROSS SECTION WITH CMS OPEN DATA

v2 after the review: MC-subtracted, closure-corrected fake factor, fiducial signal, k-fold BDT categories. 15 September 2026

$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \tau\tau, 60 < m < 120 \text{ GeV}) = 2343 \pm 39 \text{ (stat)} \pm_{-243}^{+282} \text{ (syst)} \pm 86 \text{ (acc) pb} \quad (\text{NNLO } 1945 \text{ pb})$

$\sigma_{\text{fid}}(\tau_h\tau_h) = 5.42 \pm 0.09 \pm 0.60 \text{ pb} \quad (\text{pred. } 4.50 \text{ pb}), \quad \mu_Z = 1.205 \pm_{-0.125}^{+0.145} \quad (\text{v1: } 1.160 \pm 0.192 - 0.163)$

CMS 2016 Tau dataset, Run2016G+H, 16.4 fb-1; both taus hadronic; fakes from data (same-sign fake factor); TRExFitter v1.8.0.

Precision +15/-13 % (v1: +19/-16 %): tau ID SFs 11 %, fakes 9 %, MC stat 5 %, trigger 4 %; data statistics 2.0 %.

Cross-check with DeepTau Tight on both legs: $\mu_Z = 1.071 \pm 0.114 - 0.100$ (11 % instead of 15 %), GoF p = 0.25: more precise, closer to NNLO -- recommended as the next nominal.

Sections: 1 result 2 what changed since the review 3 fit 4 BDT categories 5 fakes 6 signal definition and selection 7 findings, next steps

z-tautau/: README.md, CLAUDE.md, REVIEW.md, docs/00-09, review/ (code, documentation, review studies)

1. RESULT

NanoAOD: 160 GB data, 4 Drell-Yan + background samples



skims (trigger, $\geq 2 \tau_h$ candidates): 1.7 GB



ntuples: pair, trigger match, vetoes, masses: 330 MB (laptop)



fake factors (era $\times$ DM $\times$ N_{jets} $\times$ p_T), MC subtracted,
closure corrections $f(|\eta_1|) g(p_{T,2})$, $C_{\text{OS/SS}}$



k-fold BDT (5 folds by event number) $\rightarrow$ SR0 / SR1 / SR2



templates: fiducial signal, non-fiducial DY as background,
39 nuisance parameters + 42 γ



TRExFitter v1.8.0: fit of $m_{\tau\tau}$ in 3 categories $\rightarrow \mu_Z, \sigma$

Nominal (v2)

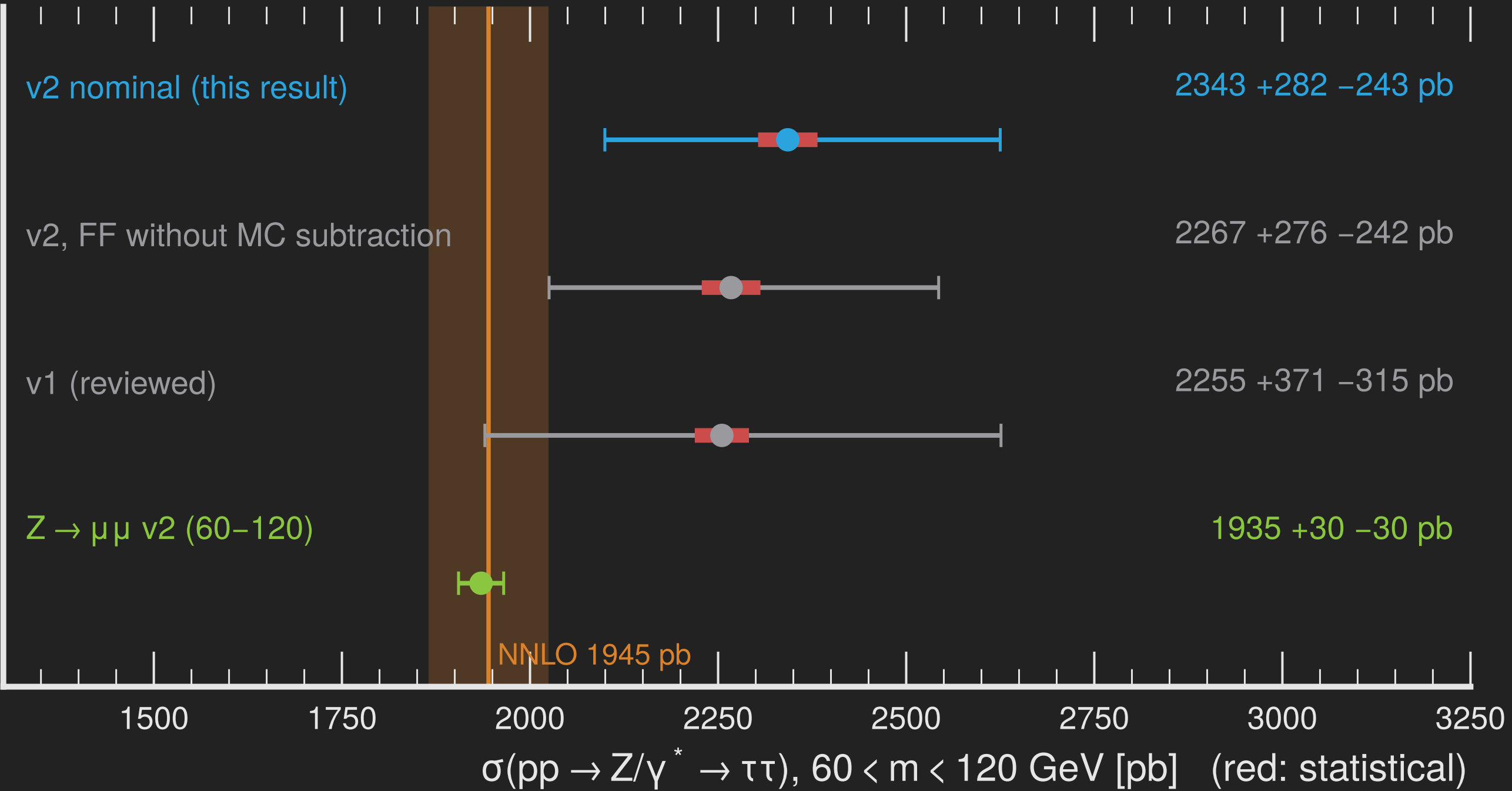
$\mu_Z = 1.205^{+0.145}_{-0.125}$ (stat. ± 0.020 , expected $^{+0.118}_{-0.102}$)
 $\sigma(60-120) = 2343^{+282}_{-243} \pm 86_A$ pb, NNLO 1945 pb
 $\sigma_{fid} = 5.42 \pm 0.09 \pm 0.60$ pb, prediction 4.50 pb
goodness of fit (saturated model): $p = 0.219$

Cross-checks

FF without MC subtraction: $\mu_Z = 1.166^{+0.142}_{-0.124}$ (fake normalisation re-measured by the fit)
v1 (single region, unsubtracted FF, whole DY as signal): $\mu_Z = 1.160^{+0.192}_{-0.163}$
 $Z \rightarrow \mu\mu$ v2: $\mu_Z = 0.990 \pm 0.014$, $\sigma(60-120) = 1935 \pm 30$ pb

Reading

1.4σ above NNLO and $\mu\mu$; the excess sits at high visible- τ p_T (section 7).
Precision limited by the external tau ID scale factors, then by the fake estimate.



Grouped impacts on

μ_z

Tau ID: 11.1 %

Fakes: 8.6 %

Gammas: 4.7 %

Tau trigger: 4.2 %

Tau energy scale: 2.1 %

Signal modelling: 1.8 %

Background normalisation: 1.8 %

MET: 1.0 %

Luminosity: 1.0 %

Pileup: 1.0 %

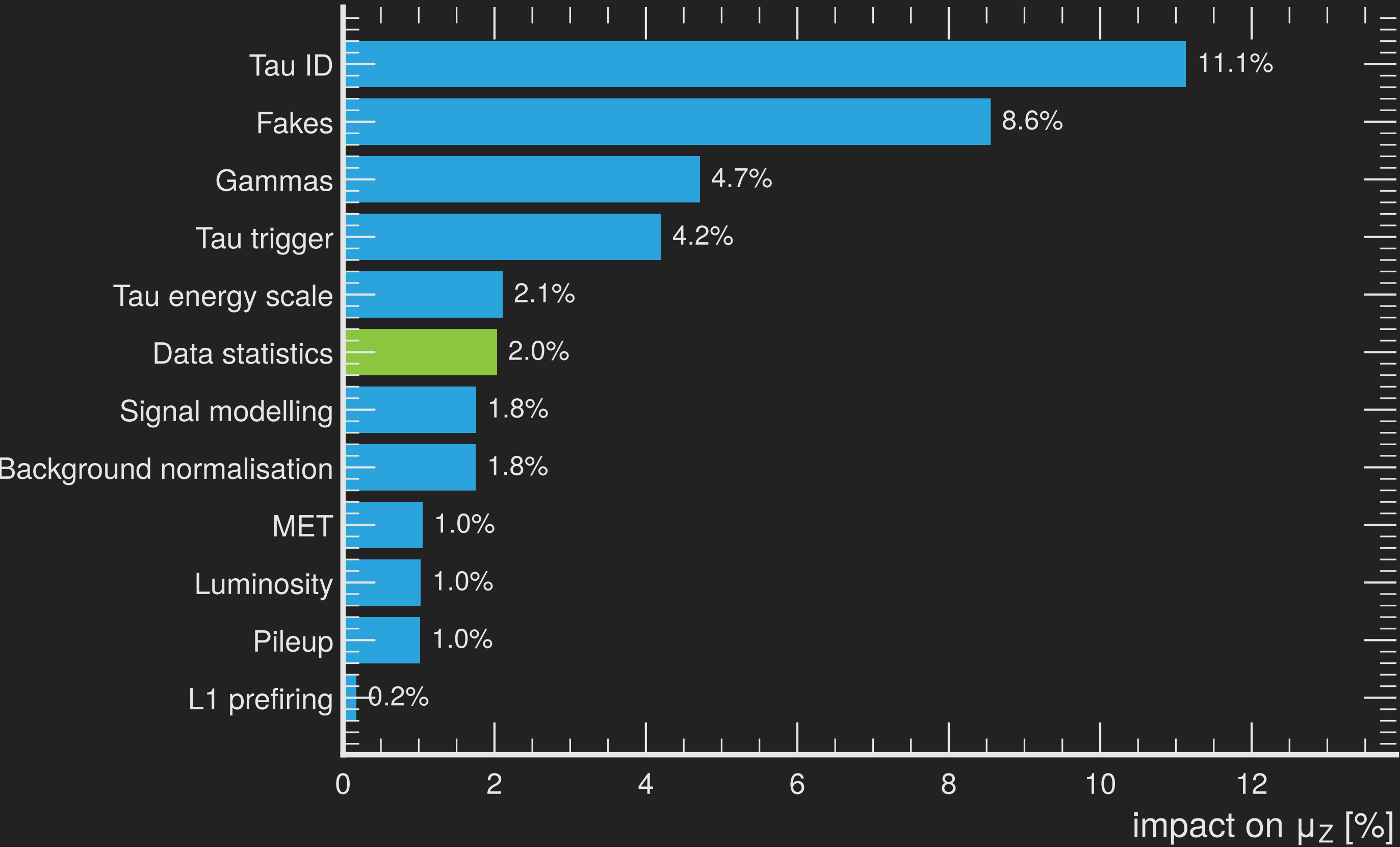
L1 prefiring: 0.2 %

data statistics: 2.0 %

v1 for comparison

tau ID 12.5 %, signal modelling 9.8 % (LO-vs-NLO NP, removed), trigger 5.1 %, MC stat 4.8 %, fakes 0.8 % (under-estimated, REVIEW.md 3.5).

Fakes now carry an honest per-category closure uncertainty; MC stat improved by the jet-binned DY samples; signal modelling is scales + PS + PDF only.



same chain (fake factors, BDT, categories, fit) with the Tight working point and its TauPOG scale factors

quantity	Medium (nominal)	Tight
mu_Z	1.205 +0.145 -0.125	1.071 +0.114 -0.100
total uncertainty on mu_Z	+14.5 / -12.5 %	+11.4 / -10.0 %
expected (Asimov)	+11.8 / -10.2 %	+10.6 / -9.3 %
data statistics	2.0 %	2.1 %
tau ID	11.1 %	8.4 %
fakes	8.6 %	6.5 %
MC statistics	4.7 %	3.9 %
tau trigger	4.2 %	3.0 %
fiducial signal / fakes (prefit, all categories)	4827 / 36985	3759 / 13592
goodness of fit p	0.219	0.250
BDT held-out AUC	0.970	0.966

2. WHAT CHANGED SINCE THE REVIEW

REVIEW.md findings (v1) and what v2 does

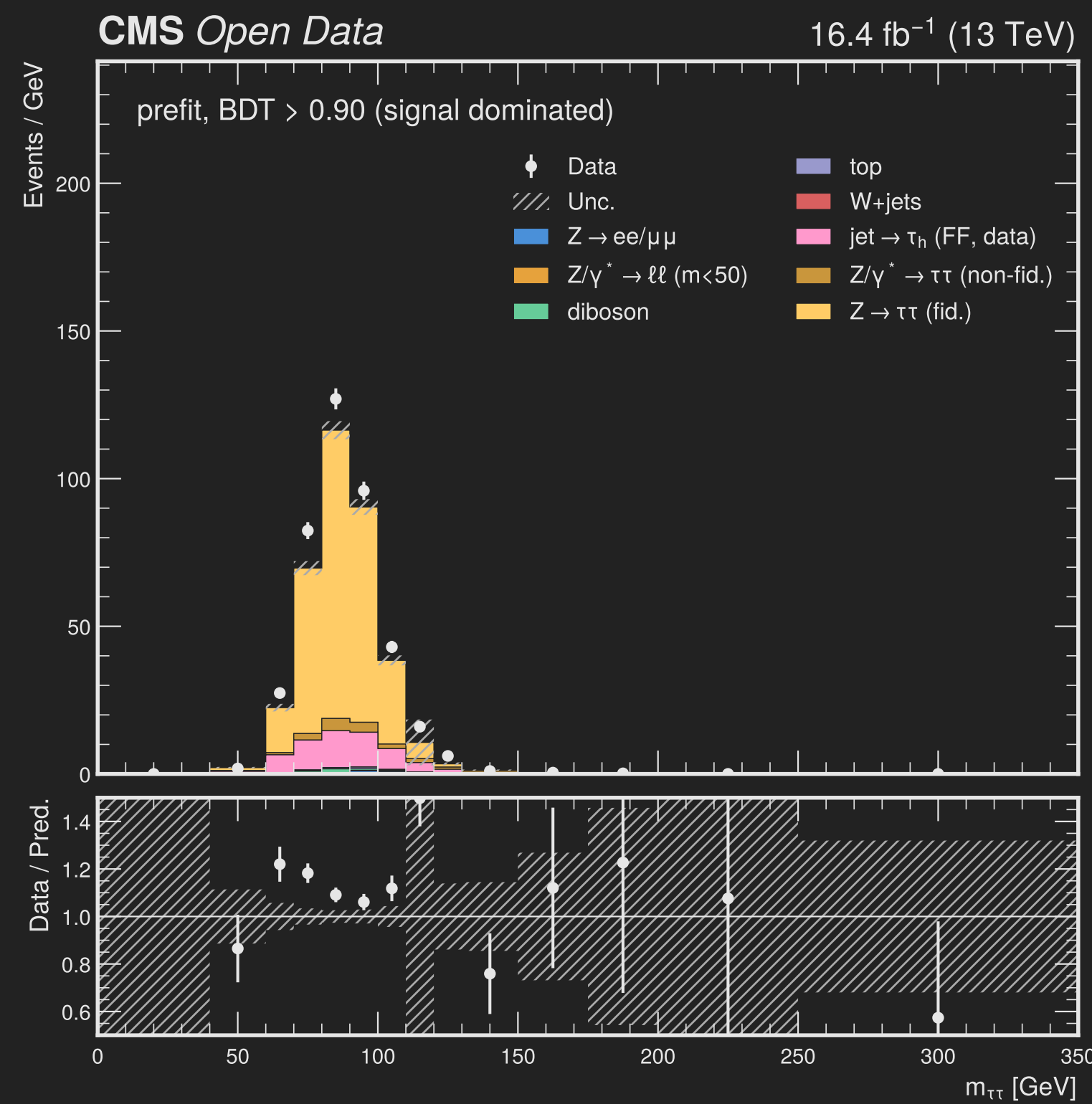
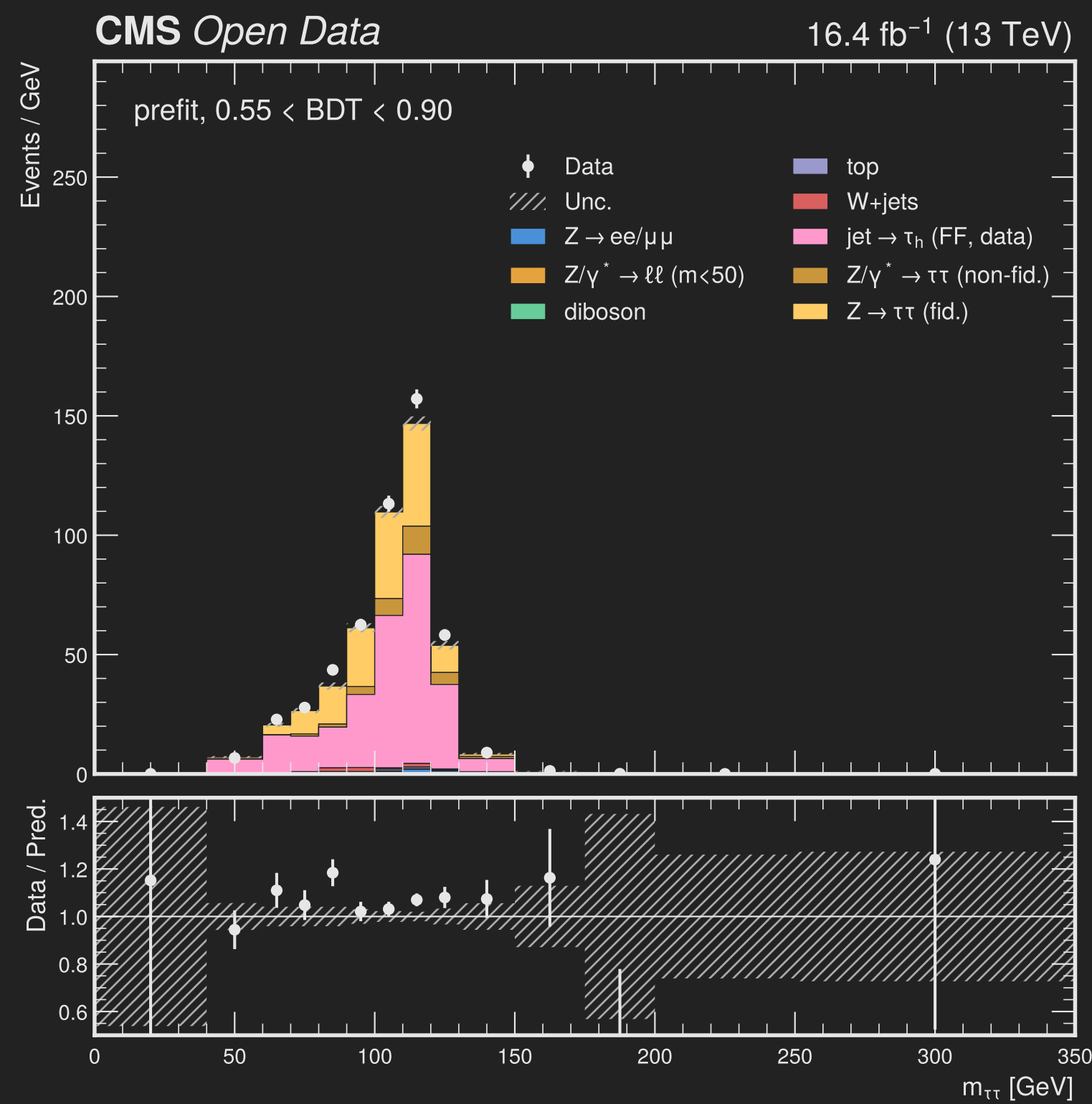
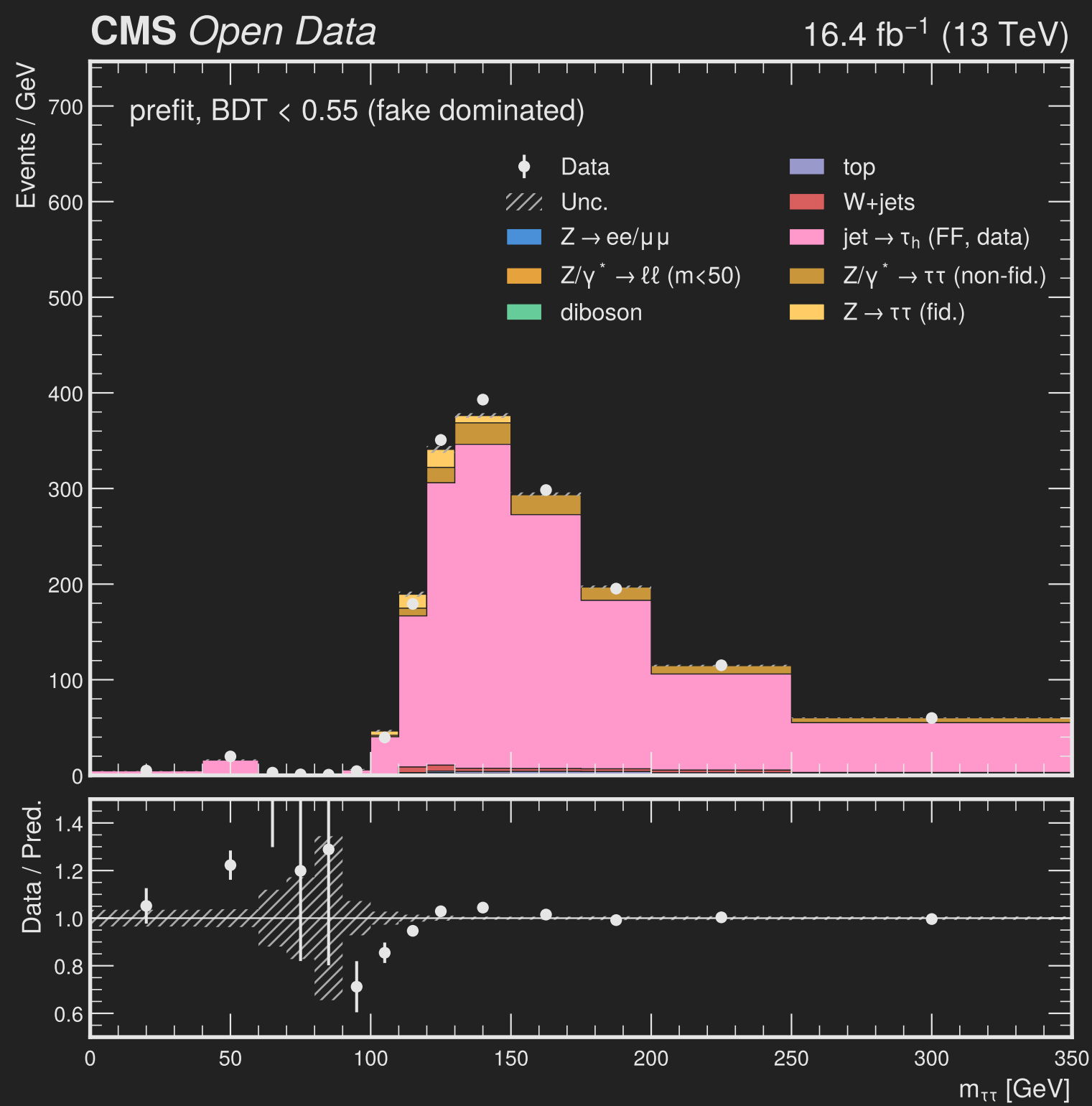
finding	v2	effect
3.1 38 % of the signal template non-fiducial	fiducial part = signal (μ_Z); rest = DYtautau_nonfid background (5 % + scales/PDF/PS on the ratio)	definition
3.2 FF without MC subtraction (bias -0.05)	MC-subtracted FF nominal; W+jets with uniform weights	+0.05 on μ
3.4 SigModel LO vs NLO 7 %, correlated with mumu	fiducial $C(\text{LO})/C(\text{NLO}) = 0.867$ reported, not fitted; NLO scales/PS/PDF	-10 % syst
3.5 one over-constrained closure NP, FF stat coherent	closure corrections $\eta(\tau_1)$, $p_T(\tau_2)$; FF stat per event; closure NPs per category x mass region	honest fakes
3.6 MC statistics 4.8 %	DY 0J/1J/2J stitched per LHE_NpNLO (5x signal MC)	4.7 %
3.8 trigger SF on jet legs	SF = 1 for genPartFlav = 0 legs	< 1 %
3.9 second TRExFitter build (StatAnalysis ROOT, v1.10)	submodule reset to v1.8.0, rebuilt with fitting/build_trexfitter.sh	reproducible
4.3 80 % fakes: ML?	5-fold XGBoost, mass-agnostic inputs, 3 categories, $m_{\tau\tau}$ fitted in each	fakes where fakes are
v2.1 fake NPs pulled 0.6-1.5 sigma	$C_{\text{OS/SS}}$ per (era, N_{jets} , BDT category); SR0 fitted only above 110 GeV (fake sideband)	pulls < 0.7 sigma
5 $\eta(\tau_1)$ non-closure +-15 %	diagnosed (FF depends on $ \eta $), corrected	η plots right

3. THE FIT: THREE BDT CATEGORIES

SR0: $BDT < 0.55$ (fake dominated; fitted above 110 GeV)

SR1: $0.55 < BDT < 0.90$

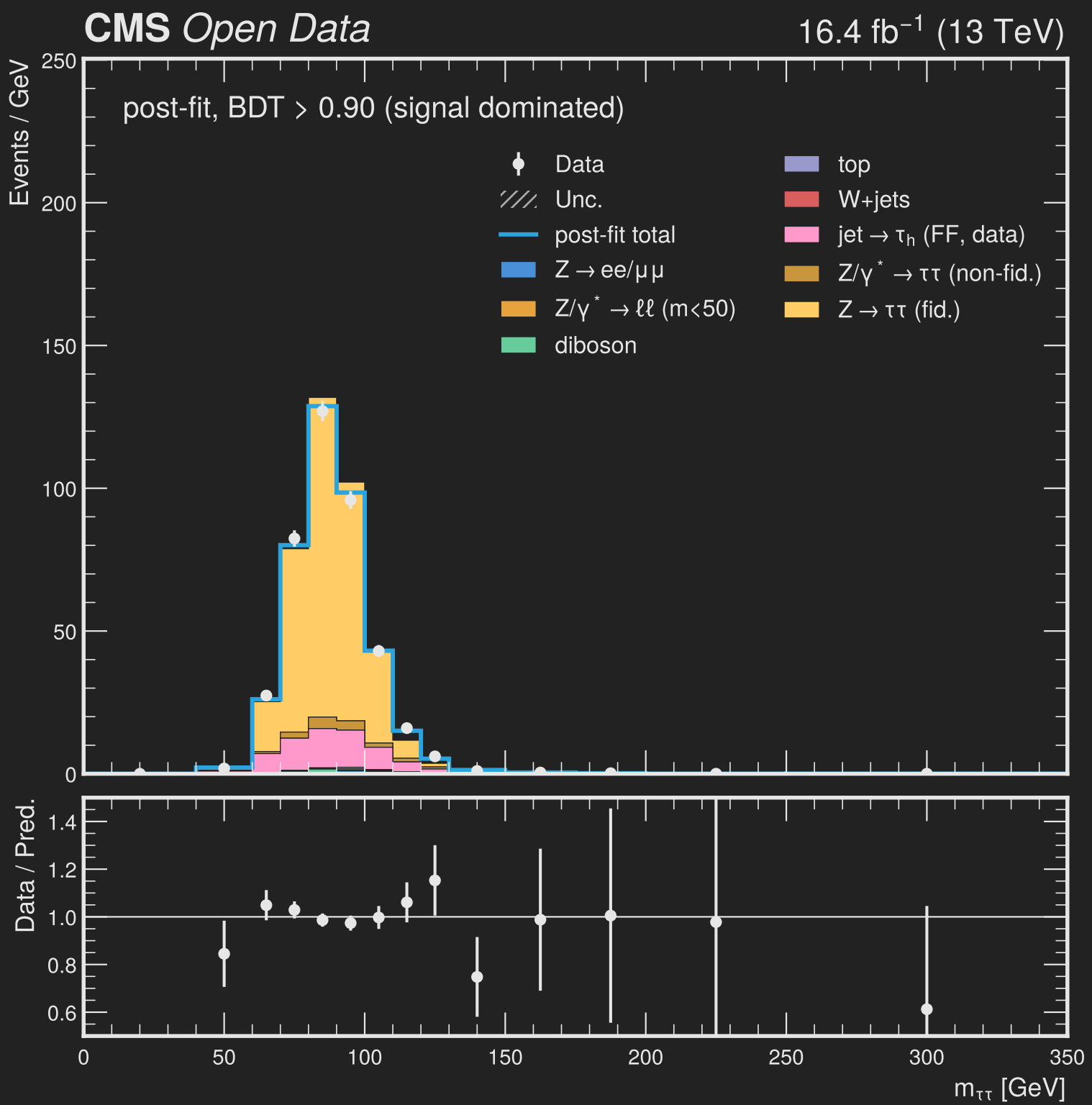
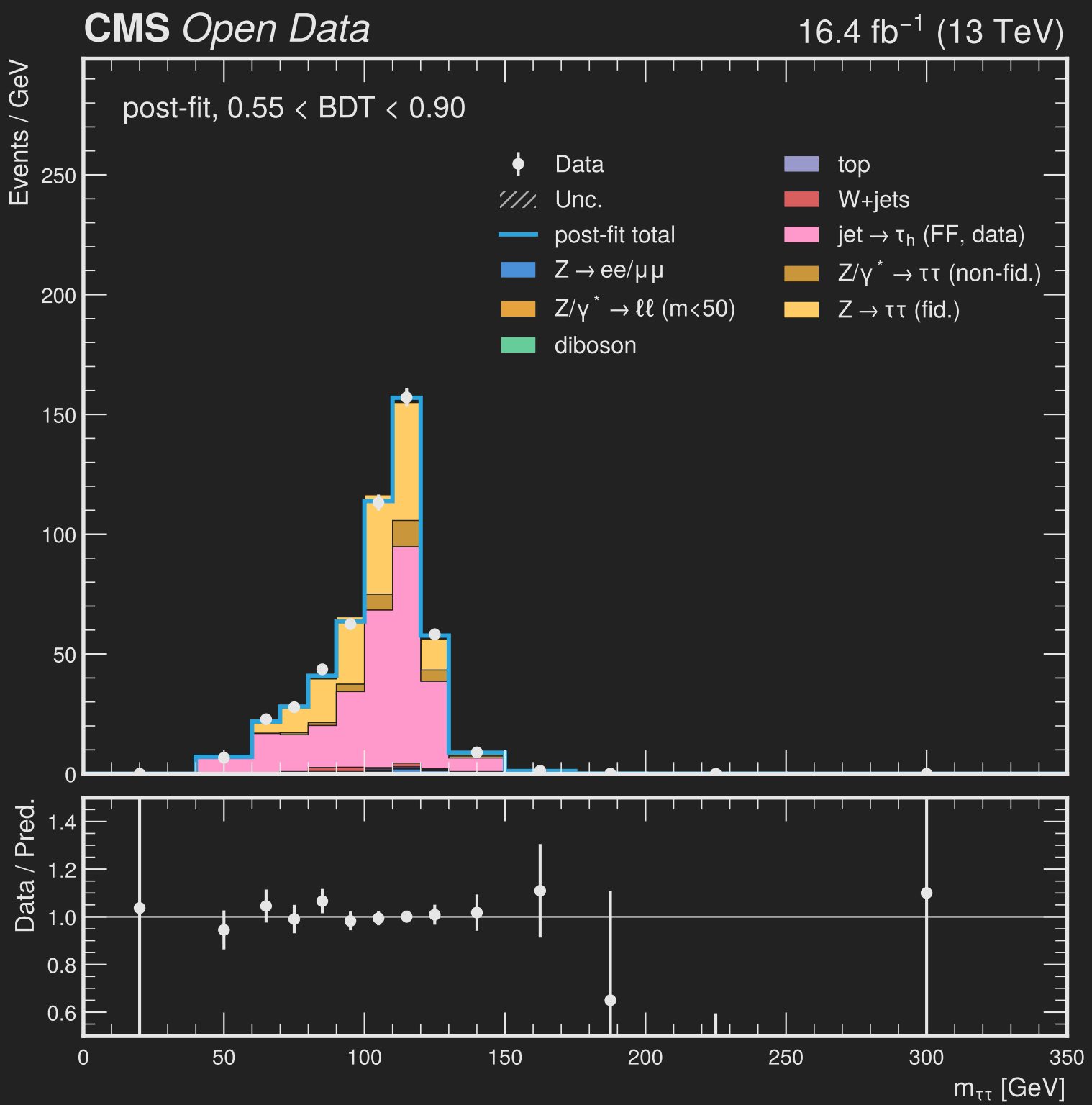
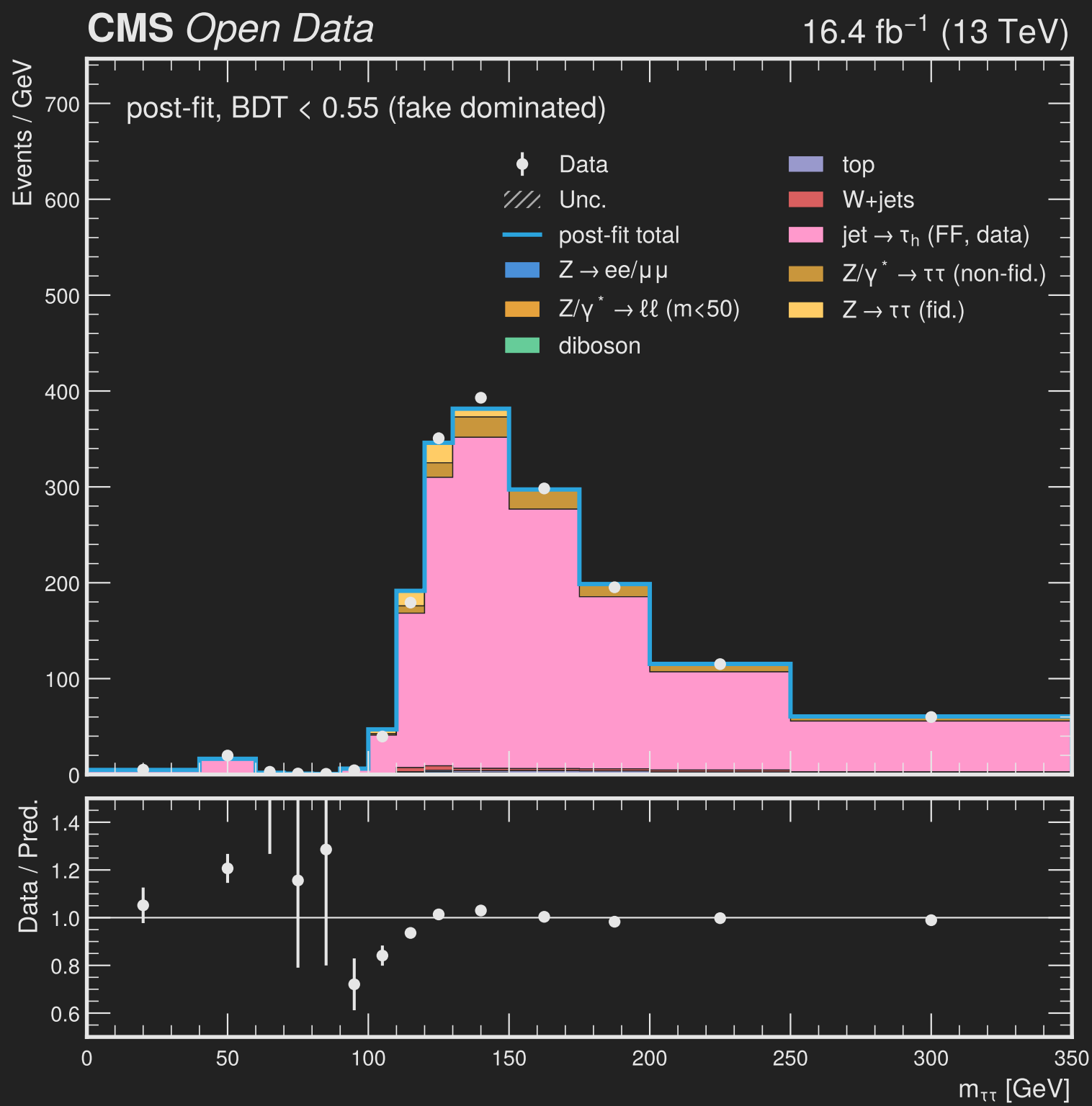
SR2: $BDT > 0.90$ (signal dominated)



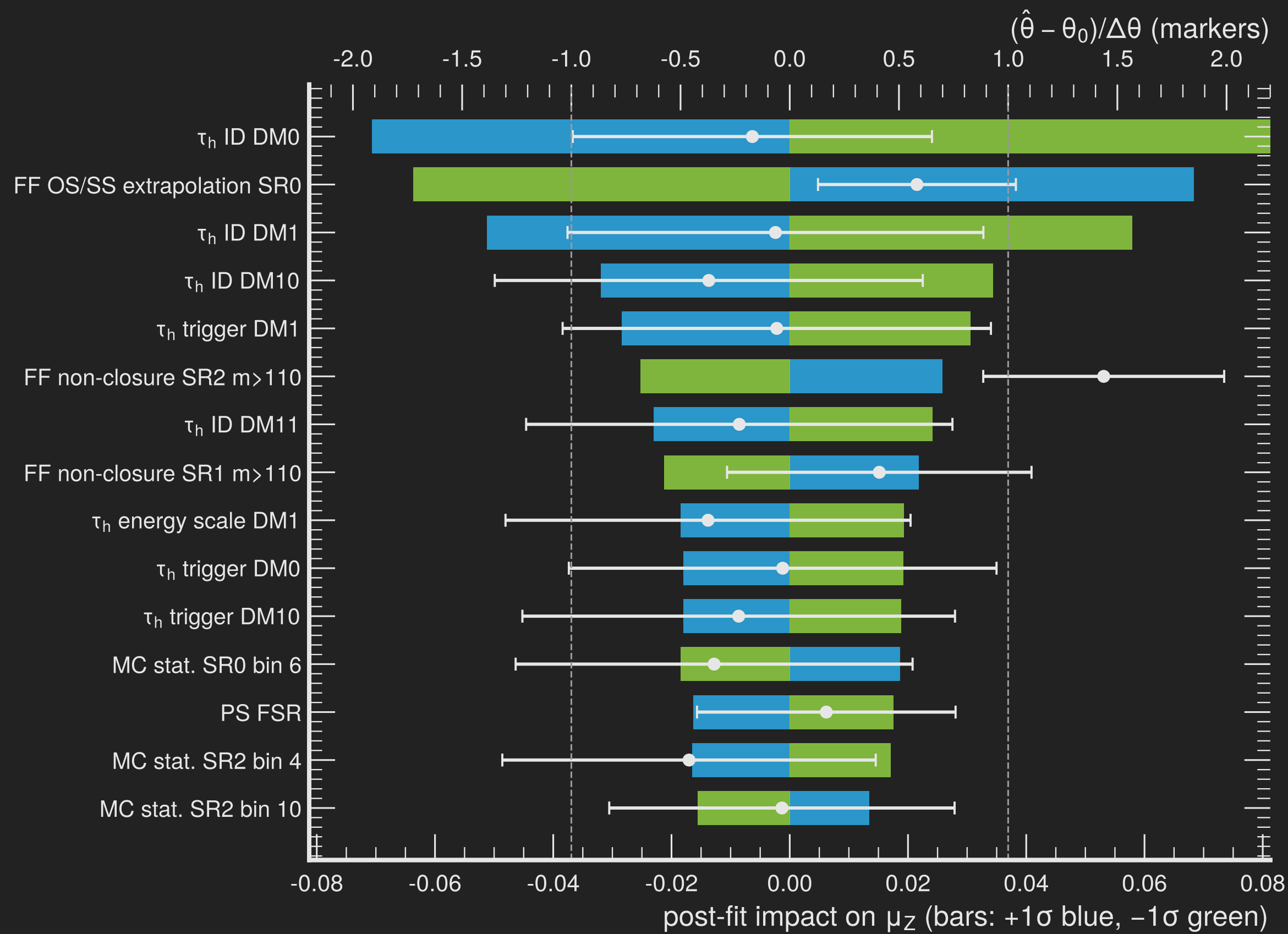
SR0 (post-fit total from TRExFitter; bins below 110 GeV not fitted)

SR1

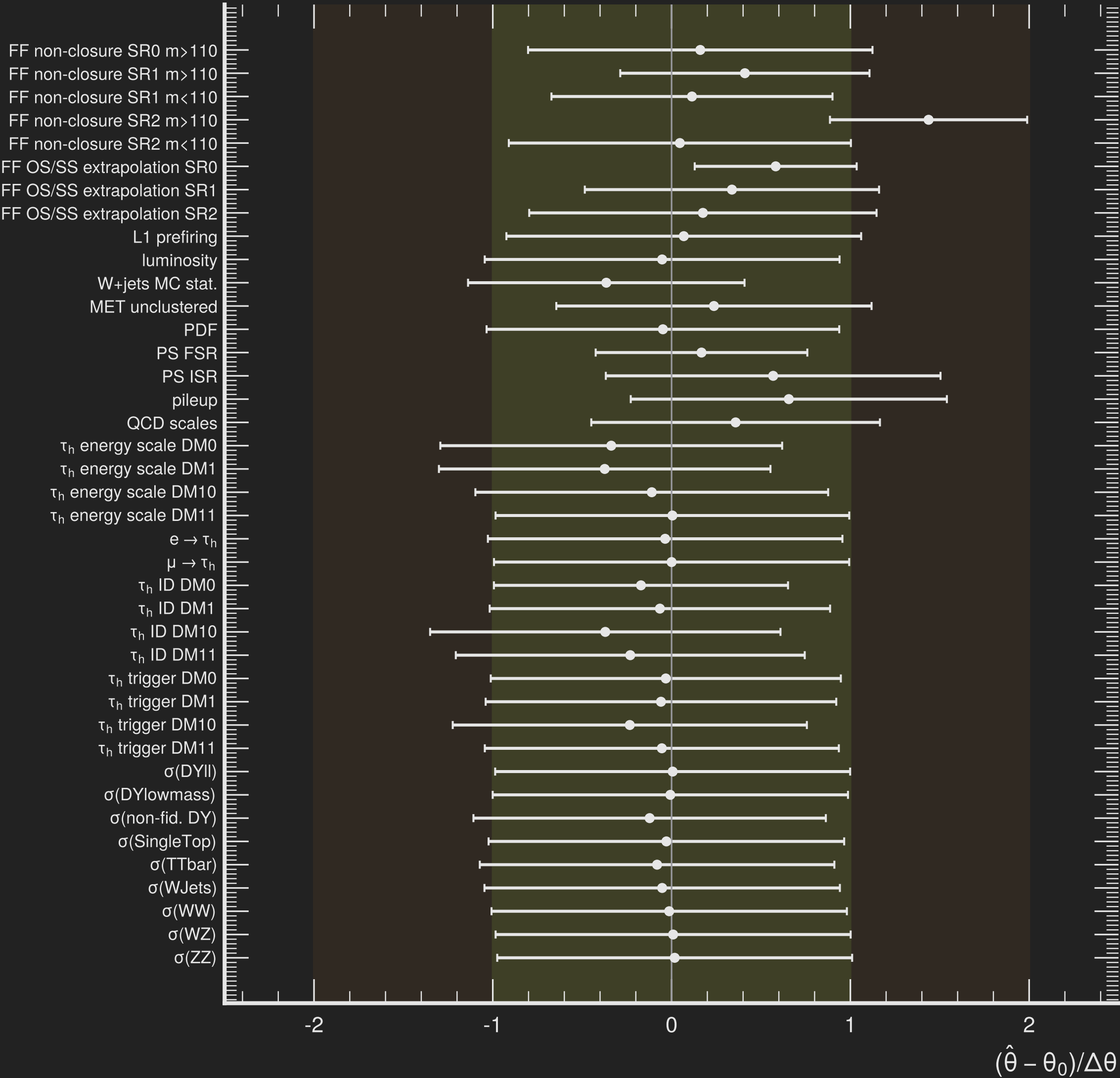
SR2: S/B = 5.1



top 15 by post-fit impact on μ_Z



all nuisance parameters



events, prefit (MC-subtracted fake factor)

sample	SR0 (< 0.55)	SR1 (0.55-0.90)	SR2 (> 0.90)	all
DYtautau	550	1487	2790	4827
DYtautau_nonfid	2529	326	169	3025
Fakes	33535	2901	548	36985
TTbar	493	52	30	575
WJets	554	63	-77	540
DYee	112	51	25	189
WW	51	3	2	56
WZ	8	7	23	38
ZZ	8	8	12	28
SingleTop	54	5	3	62
DYlowmass	20	-5	0	16
total prediction	37916	4899	3524	46339
data	38328	5204	4054	47586

4. THE K-FOLD BDT

Why

80 % fakes; every fake uncertainty scales with B/S.

Tight WP: -63 % fakes for -24 % signal, same stat. power.

A classifier keeps the signal and sorts the events.

Inputs

(mass-agnostic): $p_T(\tau_{1,2})$, ratio, $|\eta_{1,2}|$,
 ΔR , $\Delta\phi$, MET, MET significance, p_T^{vis} ,
 $\Delta\phi(MET, \tau\tau)$, $p_T(\tau\tau + MET)$, N_{jets} , jet p_T , $DM_{1,2}$

Forbidden: tau1 isolation / raw DeepTau (defines the FF regions), charge, any mass.

Training

signal = fiducial Z->tautau MC (38,645 events, inclusive + jet-binned);

background = the fake estimate itself (AR data x FF, 187,973 events).

XGBoost, depth 4, 300 trees, both classes with equal total weight.

5 folds by event number

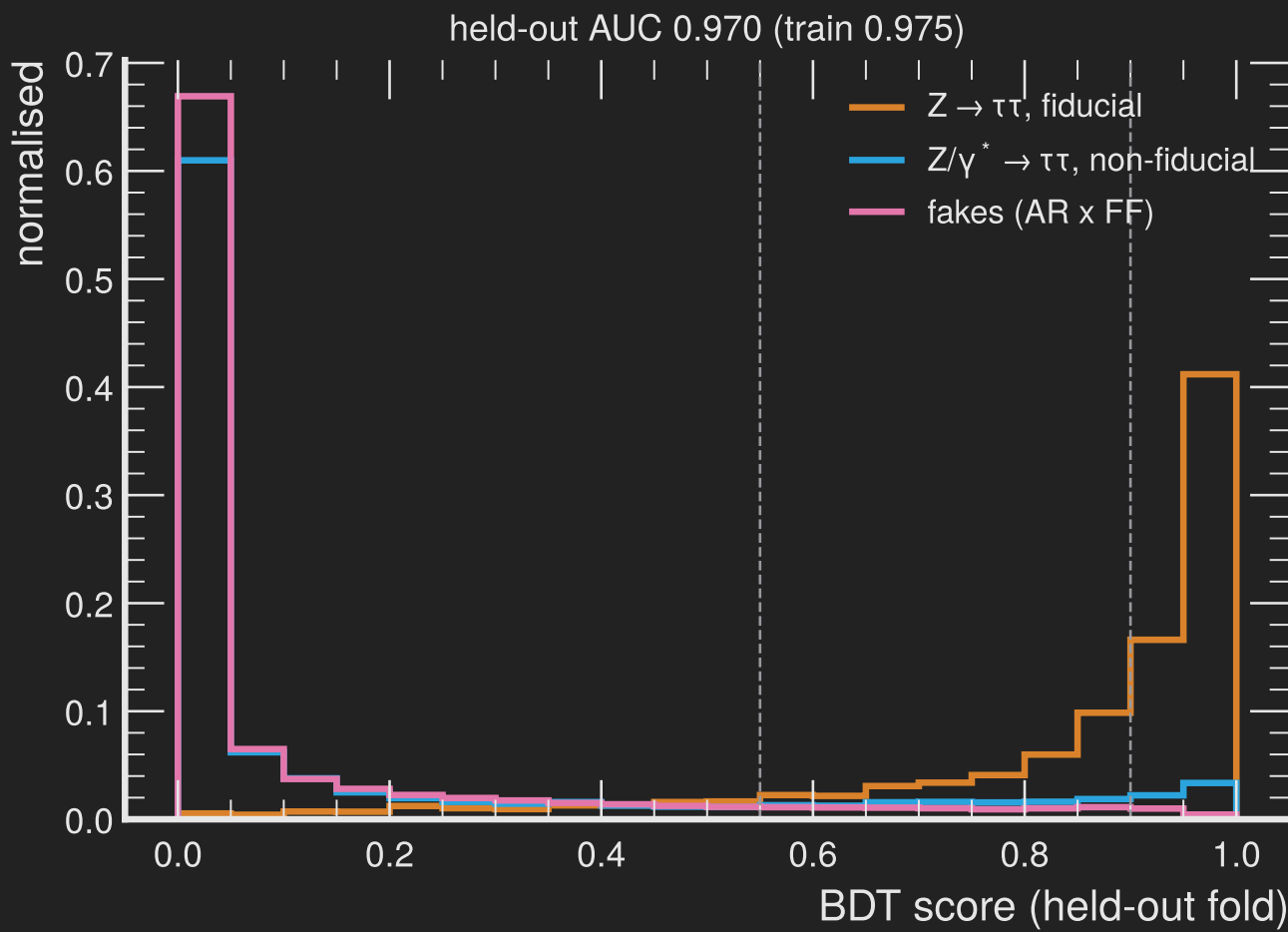
AR data, same-sign data, SR data, every
MC sample and every kinematic variation are scored by the model
that never saw them. Held-out AUC 0.970 (train 0.975).

Categories

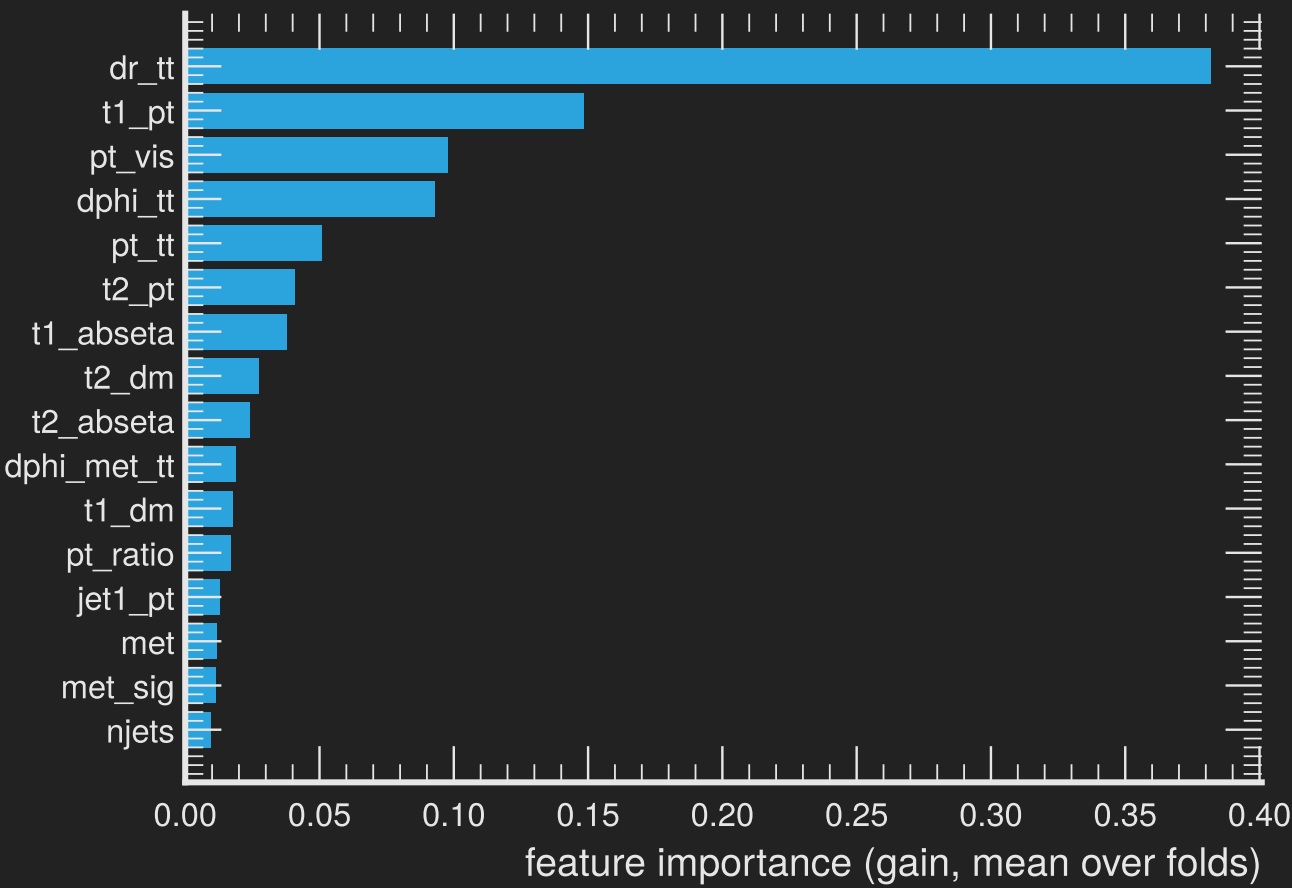
< 0.55 (fake dominated), 0.55-0.90, > 0.90 (S/B 5.1).

Stat-only sensitivity 1.97 % -> 1.84 % (fakes fixed).

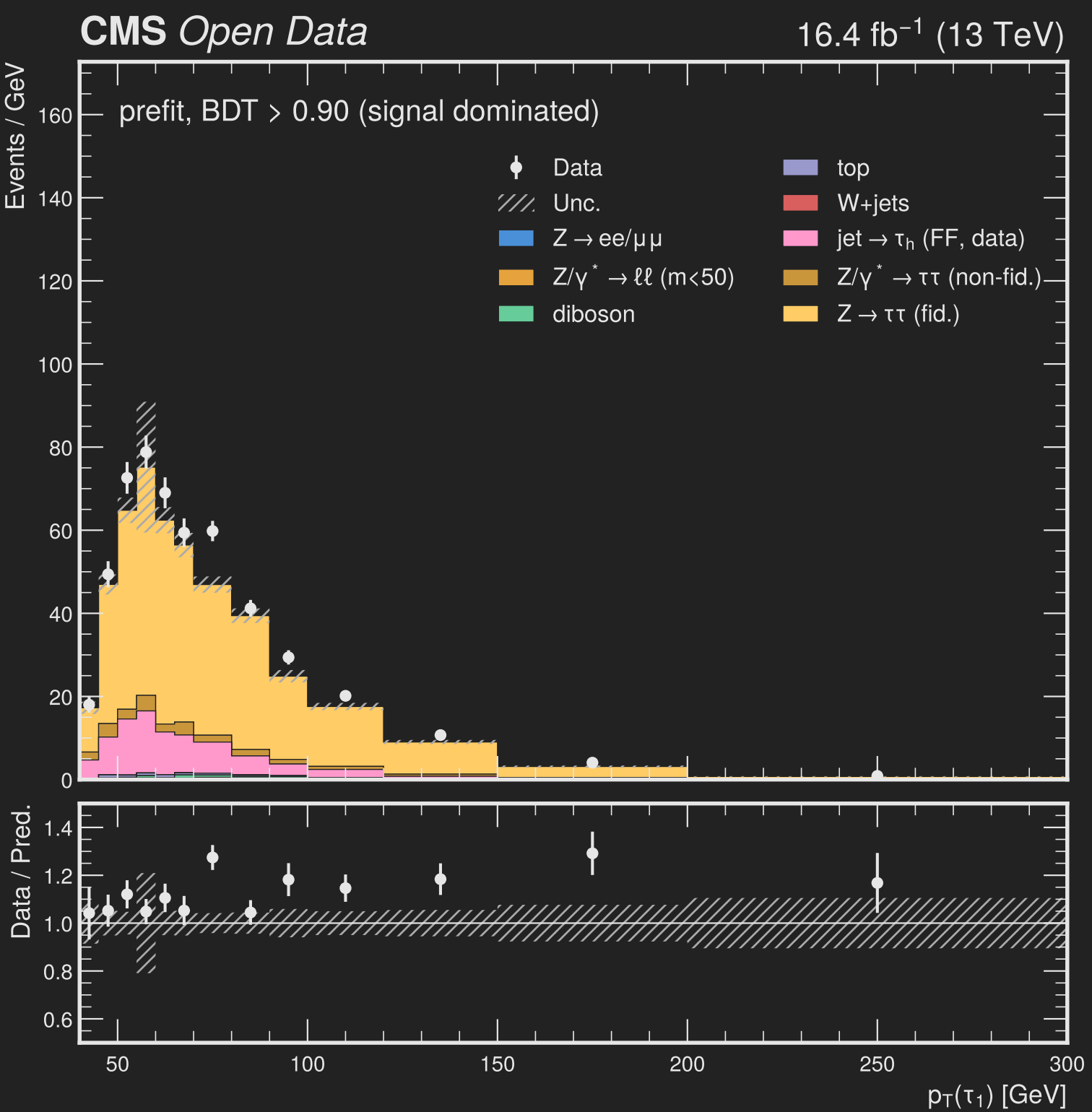
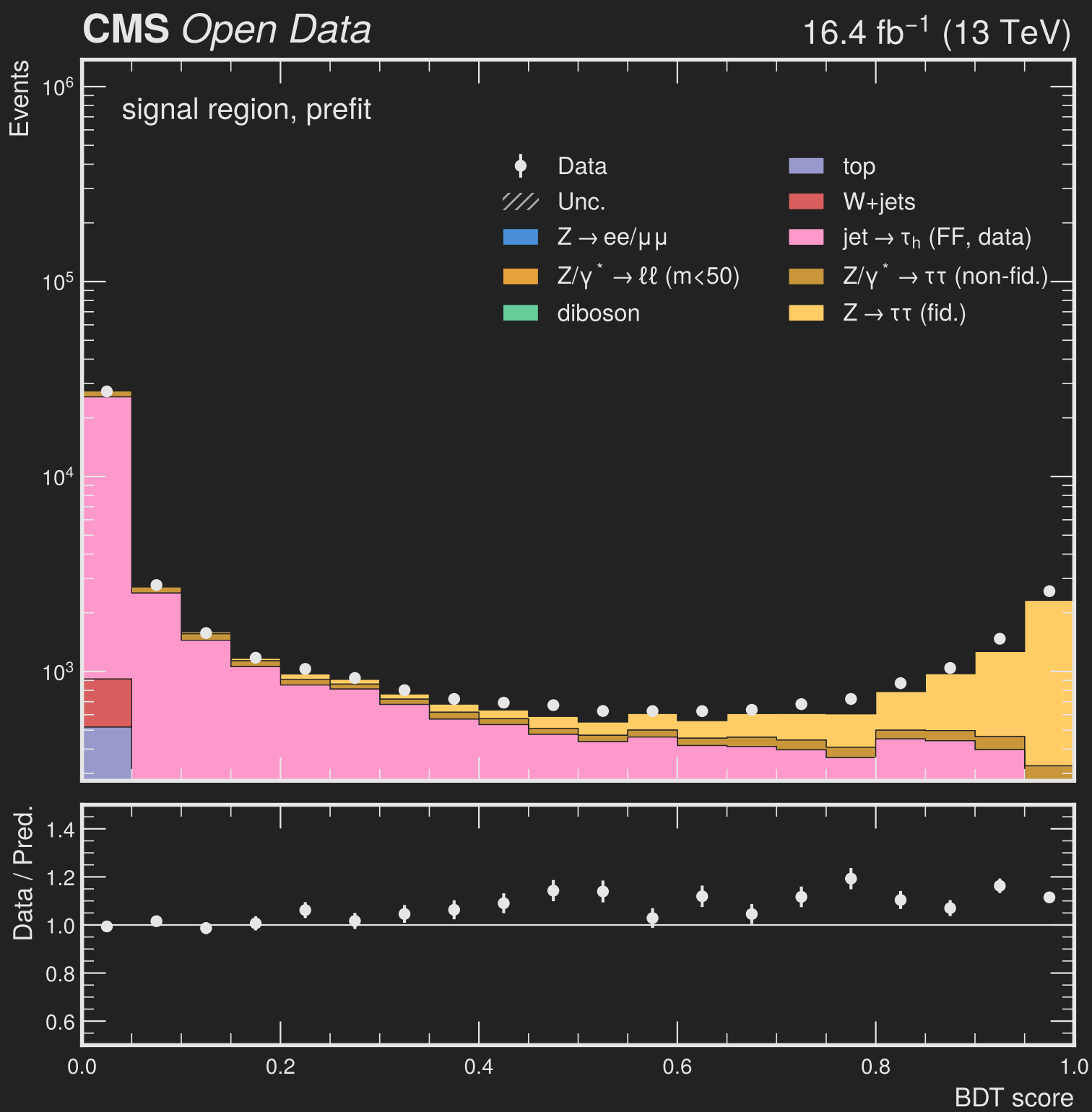
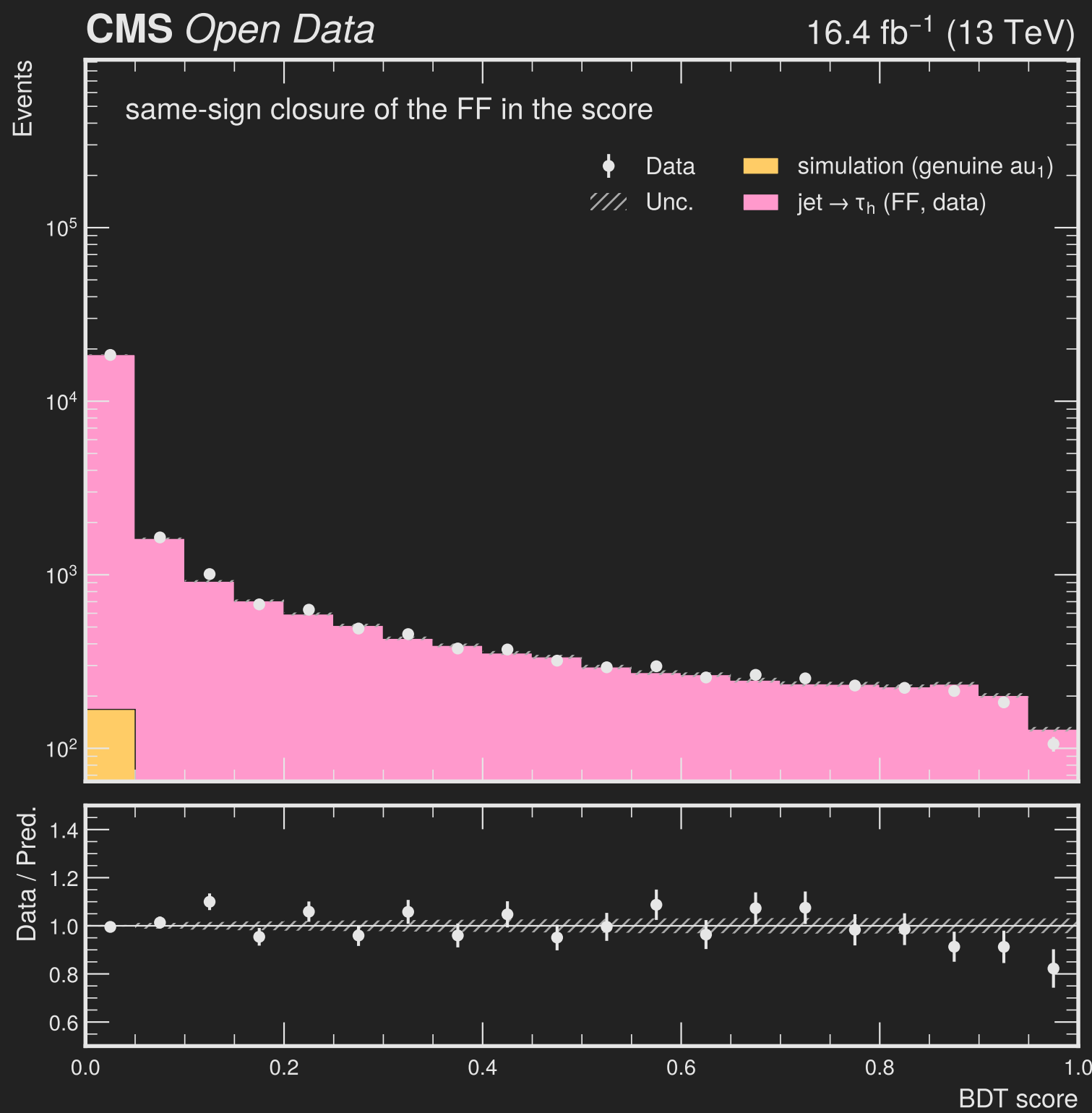
score shapes (held-out folds)



feature importance



same-sign closure in the score: SS_T vs $FF \times SS_L$ (MC subtracted) region, prefit: the fake-dominated bins close, the excess is ~ 2
SR2: $p_T(\tau_1)$ data vs prediction (the trend behind section 7)



$FF = [N(SS, \tau_1 M, \tau_2 M) - MC] / [N(SS, \tau_1 VVVL\Lambda \neg M, \tau_2 M) - MC]$
 $\times f(|\eta_1|) g(p_{T,2})$; applied to OS events with τ_1 failing Medium,
times $C_{OS/SS}(era, N_{jets})$ from the τ_2 -anti-isolated sideband.

era x DM x N_jets x pT = 120 bins, ~10 % stat. each (per event into the template variance).

MC subtraction

genuine tau1 = 1.3 % of the AR but 6 % of the signal,
3.9 % of the C numerator; C: 1.078 -> 1.052.

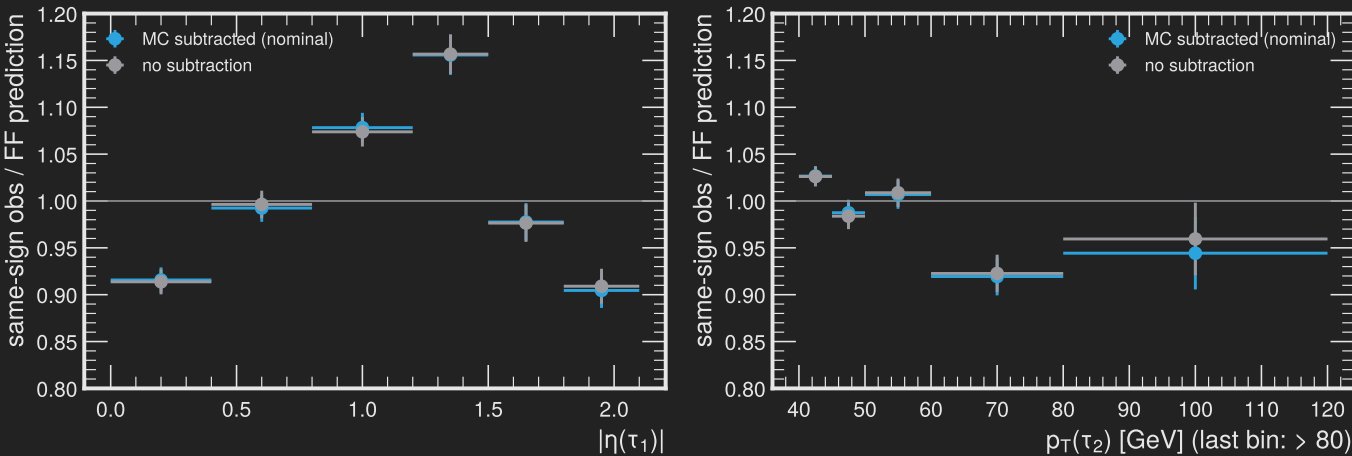
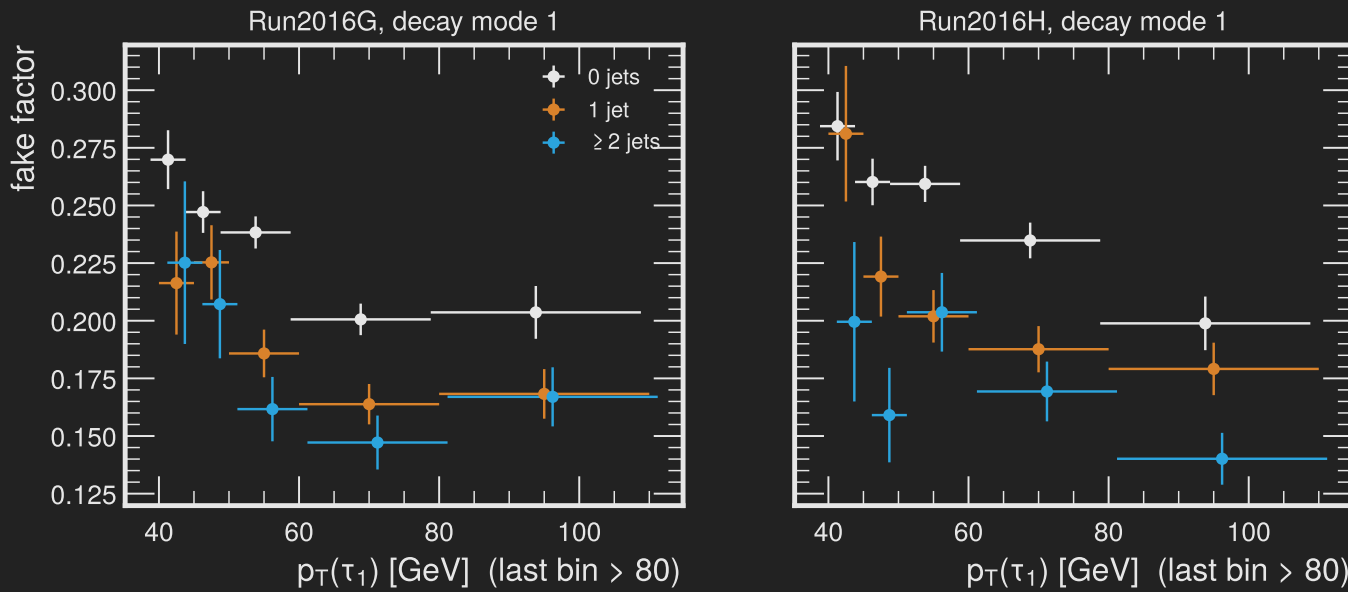
W+jets subtracted with uniform weights (raw weights 20-200, one at -63).

Closure corrections

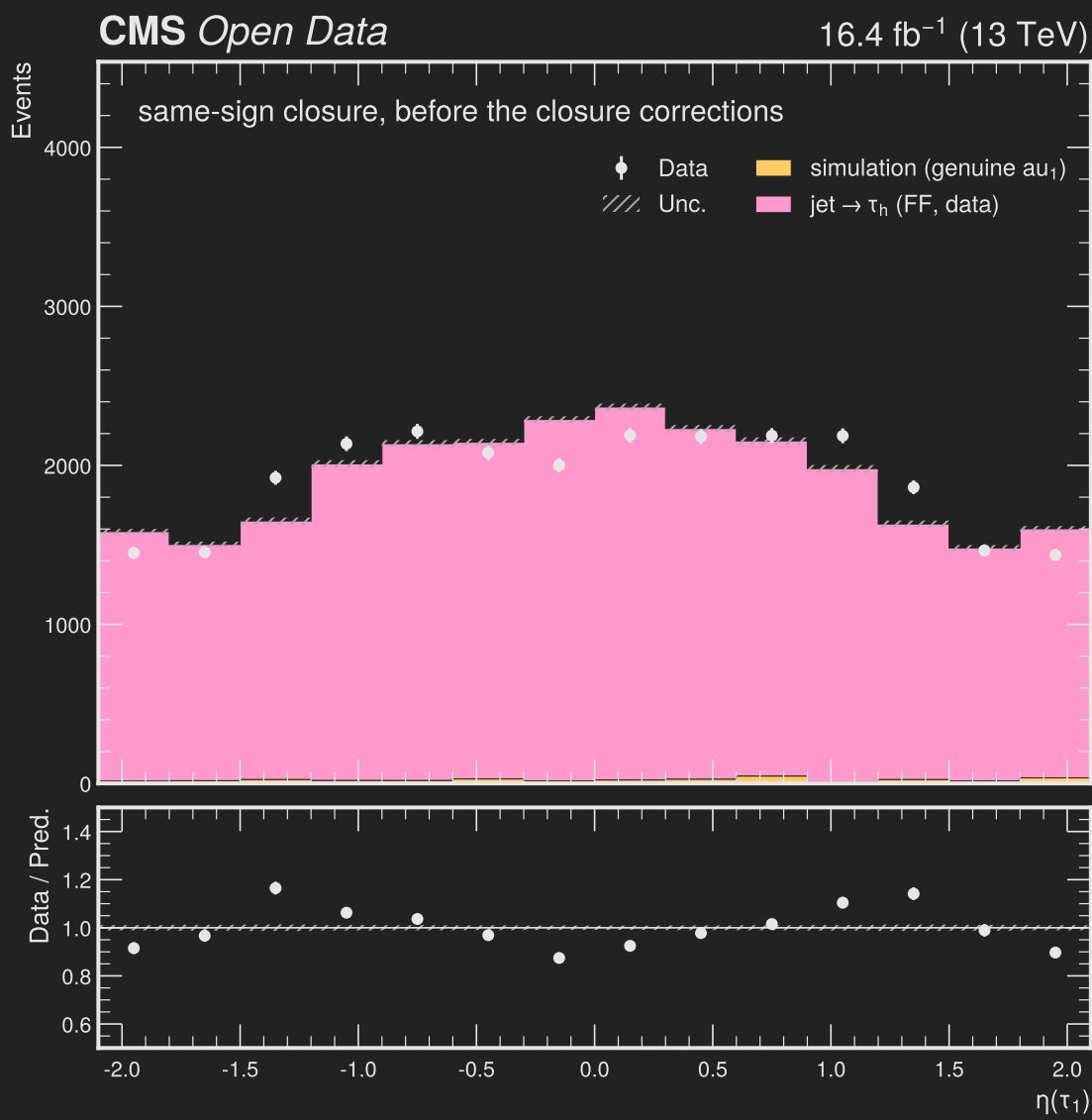
the FF varies by +-15 % with $|\eta(\tau_1)|$ (same shape
in both eras, all pT, N_jets and DMs: DeepTau's jet rejection is not eta-flat)
and by -6 % with pT(tau2). Residual non-closure per category and mass
region -> one nuisance parameter each: c0/lo 15 %, c0/hi 1 %, c1/lo 5 %, c1/hi 7 %,
c2/lo 12 %, c2/hi 55 %.

fake factors, decay mode 1

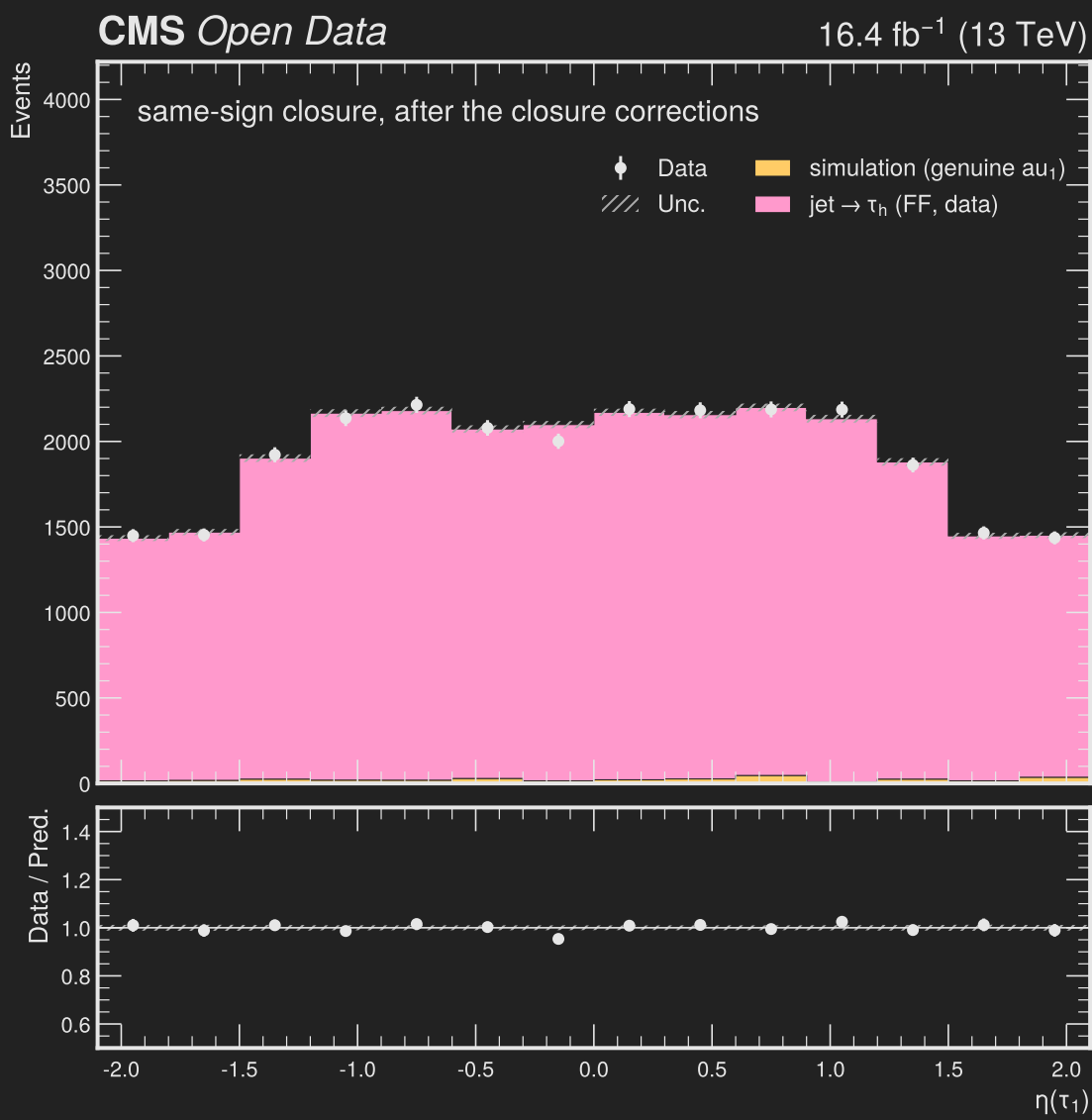
closure corrections



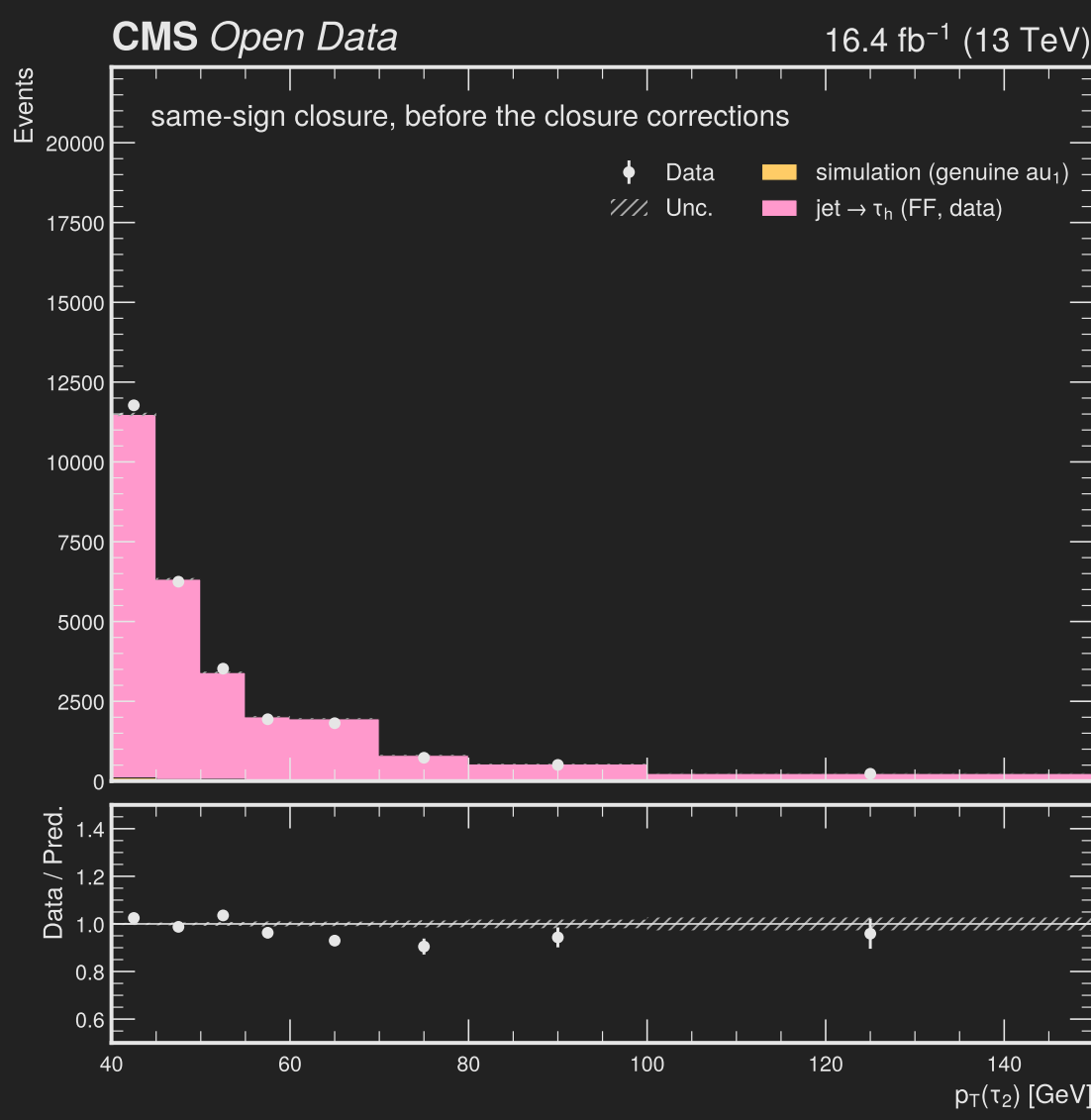
$\eta(\tau_1)$ before



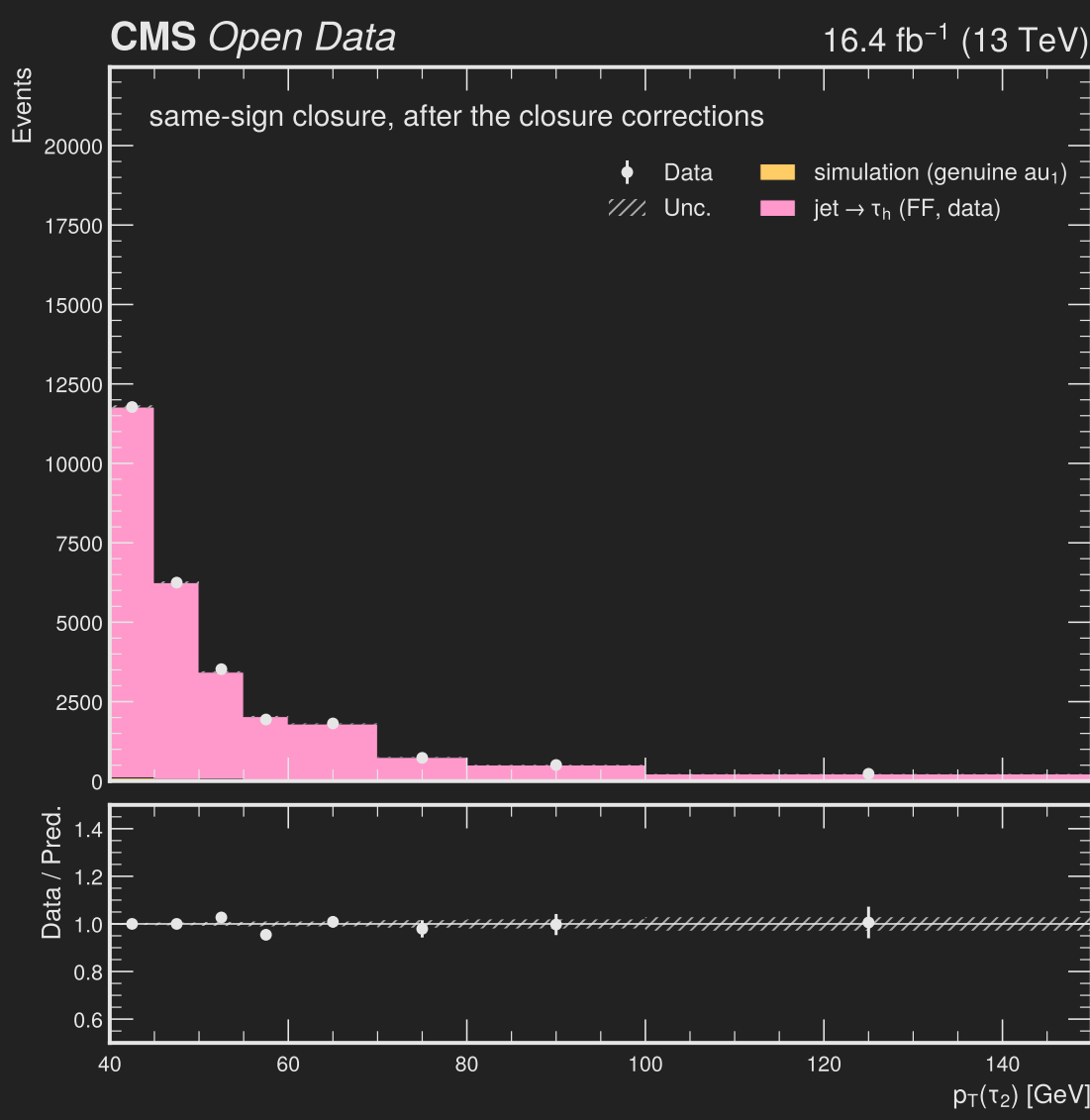
$\eta(\tau_1)$ after

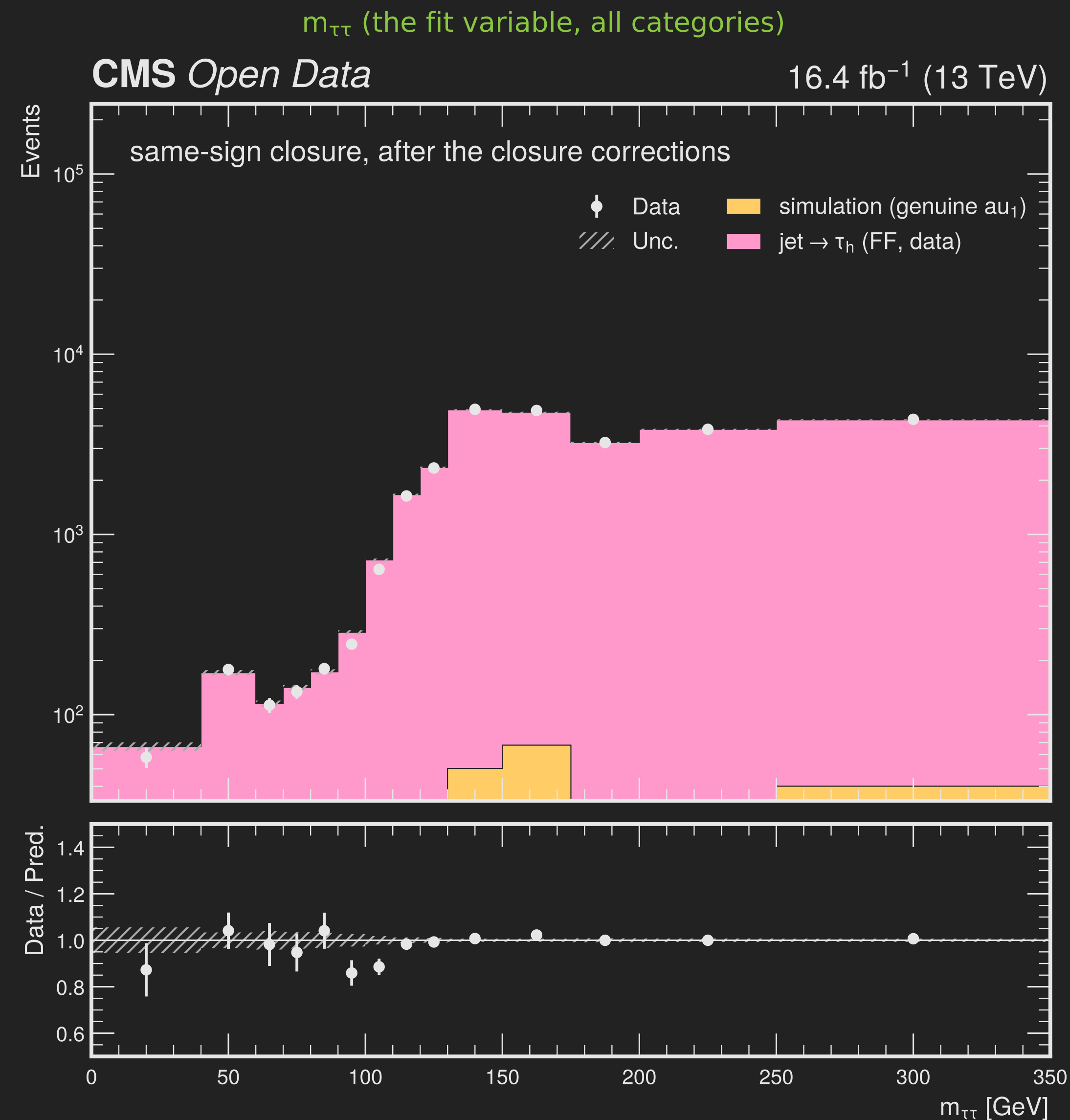
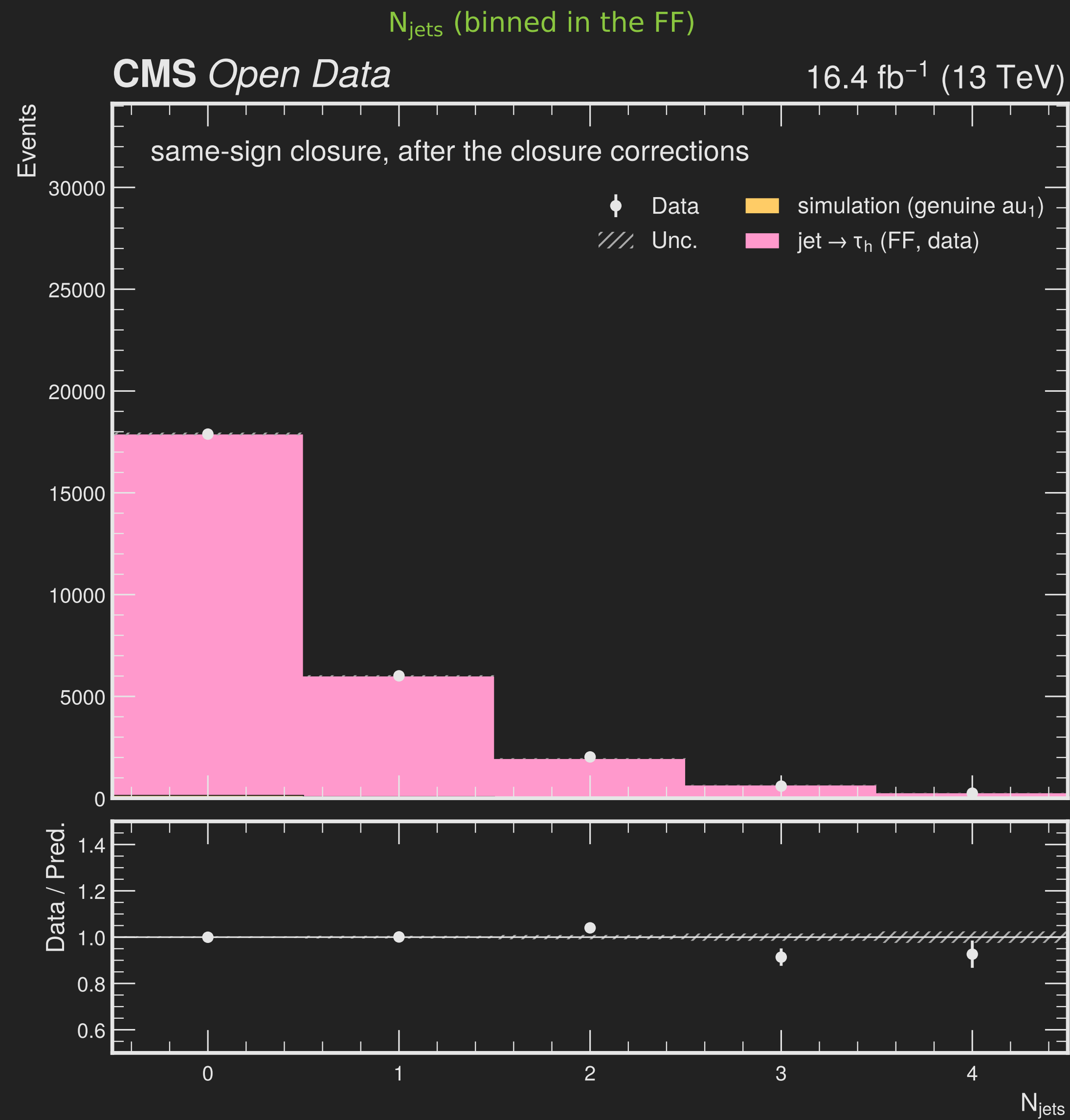


$p_T(\tau_2)$ before



$p_T(\tau_2)$ after





6. SIGNAL DEFINITION, MASS, SELECTION

Fiducial volume

60 < m_LHE < 120 GeV, both taus hadronic, both visible taus
pT > 40 GeV, |eta| < 2.1. A = 0.231 % (+-3.7 %), sigma_fid(pred) = 4.50 pb.

Of the selected Z/gamma* -> tautau

62 % fiducial (-> DYtautau, x mu_Z),
30 % has m_LHE > 120 GeV, 6 % migrates across the 40 GeV / |eta| cuts, 1.5 % has a tau->e/mu leg.

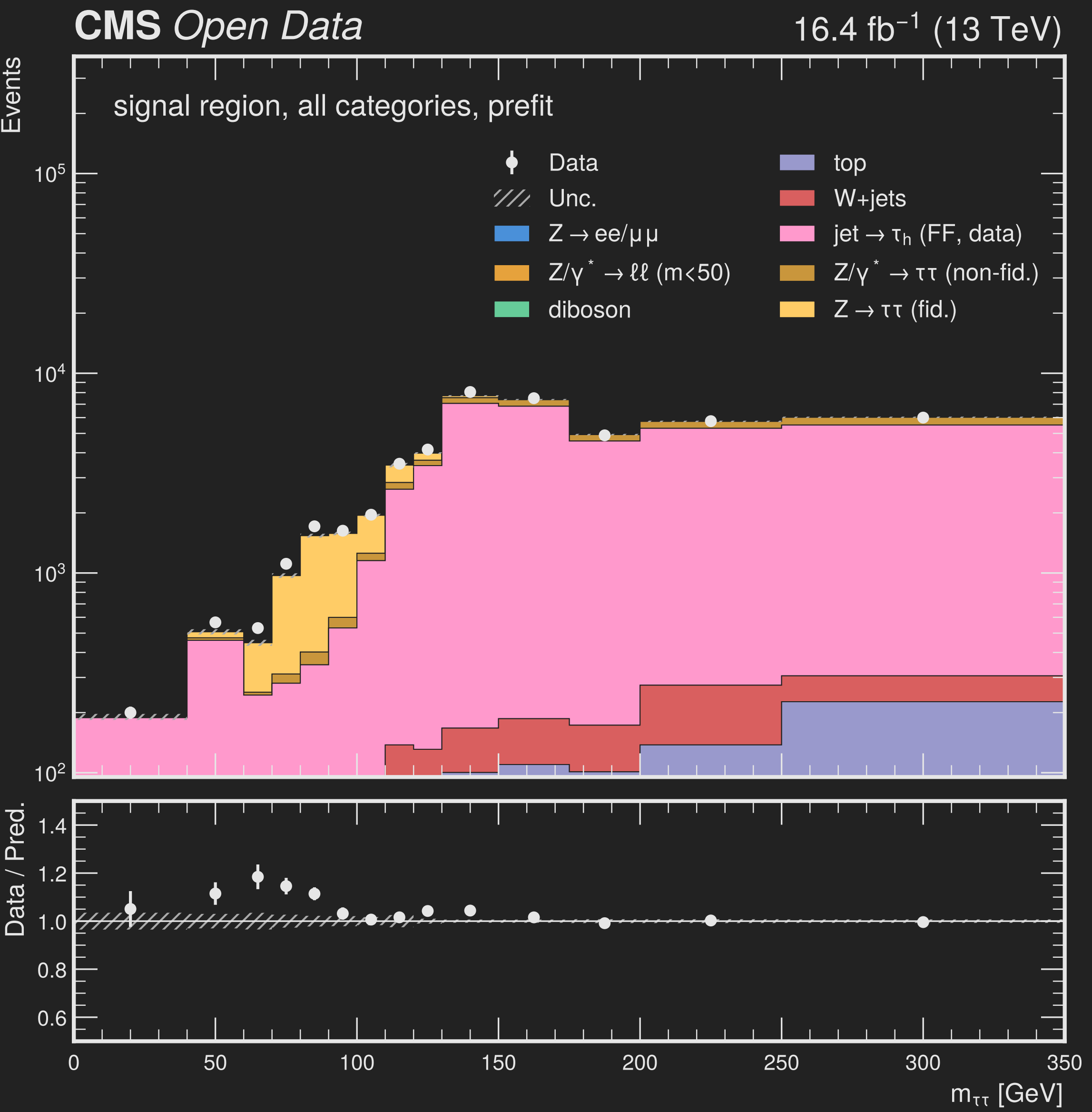
The two 40 GeV cuts enrich the gamma* continuum by two orders of magnitude: 88 % of the Z->tautau above m = 130 GeV (the fake sideband) has m_LHE > 120 GeV.

v2

DYtautau_nonfid is a background (5 % normalisation NP + the scale/PDF/PS variations acting on the non-fid/fid ratio, all experimental NPs correlated).
sigma_fid = mu x sigma_fid(pred) is literally true; the sideband no longer contains mu-dependent signal (REVIEW.md 3.1).

Generator cross-check

fiducial C(LO madgraph)/C(NLO) = 0.867 -- the LO sample's softer visible-tau spectrum meets the trigger turn-on. Reported, not an NP (docs/07).



≥ 4 neutrinos escape: m_{vis} peaks at $0.8\ m_Z$.

Collinear approximation

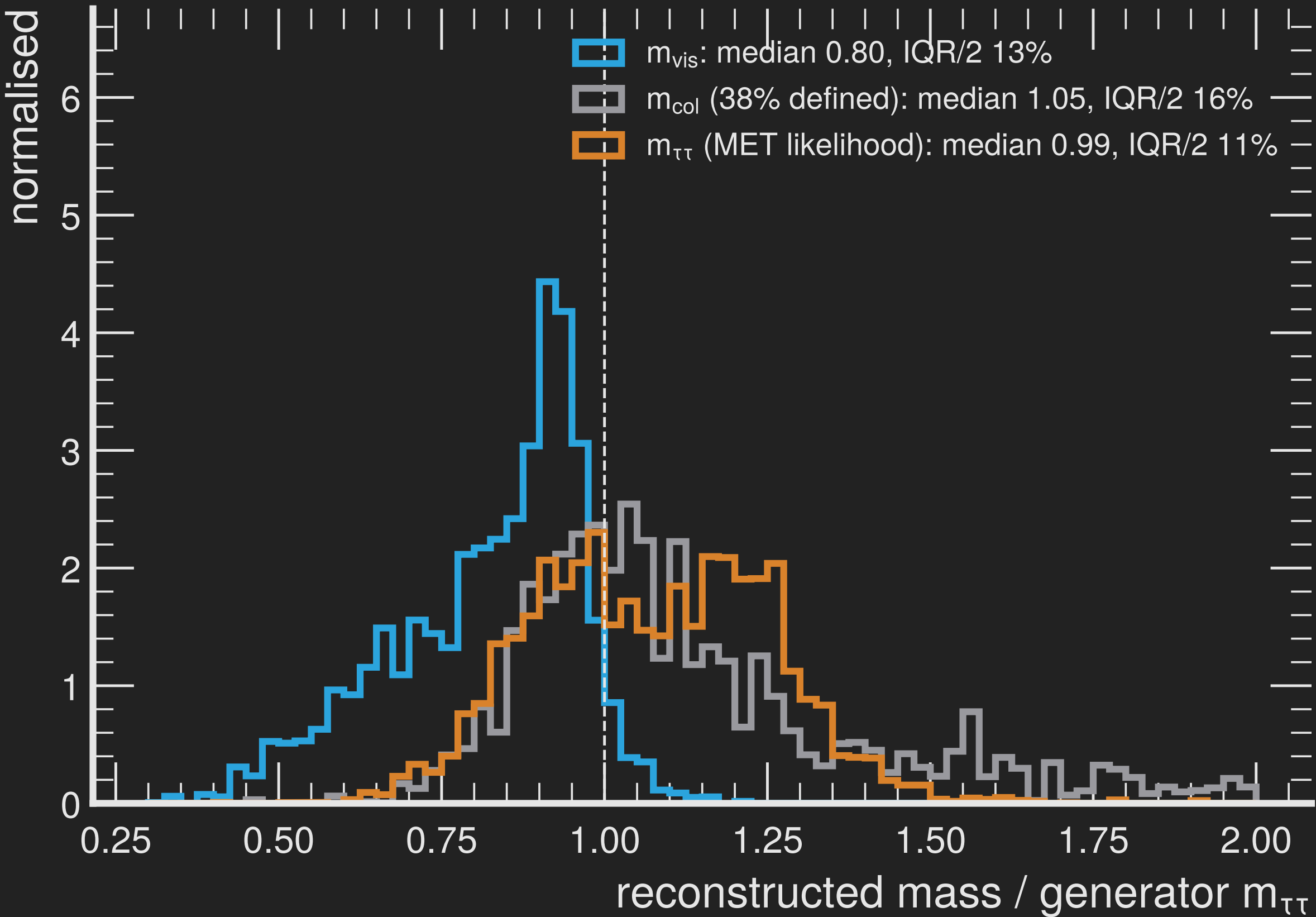
singular for back-to-back taus, defined in 38 %.

Likelihood mass

(FastMTT / SVfit-lite idea): scan the visible fractions
 (x_1, x_2) , weight = MET transfer function (per-event covariance)
 $\times$ flat two-body phase space $\times 1/m^2$; posterior median.

scale 0.99, resolution 11 % (m_{vis} : 0.80, 13 %); always defined.

Separation from fakes about equal to m_{vis} ; $m_{\tau\tau}$ is the physical mass
and gives a clean high-mass sideband for the fake normalisation.



selection

object	requirement
trigger	DoubleMedium(Combined)IsoPFTau35_Trk1_eta2p1_Reg (G: Iso, H: CombinedIso); both legs matched to HLT tau objects
tau_h	pT > 40 GeV, eta < 2.1, dz < 0.2 cm, DM 0/1/10/11; DeepTau VSe VVLoose, VSmu VLoose; VSjet Medium (tight) / VVVLoose & !Medium (loose)
pair	most isolated pair, dR > 0.5; tau1 = leading pT (the fake-factor leg)
veto	no extra e / mu: orthogonal to e tau_h, mu tau_h, ee, mumu
simulation	events whose leading tau is a jet are dropped (covered by the fake factor)

cutflow

step	Run2016G	Run2016H
NanoAOD	79.6 M	76.8 M
skim (certified, trigger, >= 2 candidates)	2.81 M	2.22 M
pair + MET filters	2.03 M	1.62 M
trigger match	1.99 M	1.59 M
lepton veto	1.99 M	1.59 M
signal region (OS, both Medium)	24 106	23 480

7. FINDINGS AND NEXT STEPS

1. $\sigma(60-120) = 2343^{+282}_{-243} \pm 86$ pb, $\mu_Z = 1.20$: 1.4σ above NNLO and above $Z \rightarrow \mu\mu$ ($\mu_Z = 0.99$). Total $^{+15}_{-13}$ % (v1: +19/-16 %), data statistics 2.0 %.
2. The limit is external: the TauPOG tau ID scale factors (11 %); Z->tautau is how those SFs are measured, and the two POG prescriptions (DM- vs pT-binned) differ by 14 % on the yield.
3. In the signal-dominated category data/prediction rises with the visible tau pT (1.0 at threshold, 1.2 above 70 GeV): part of $\mu > 1$ comes from there (the fit quality itself is fine, $p = 0.22$, once the fake sideband is restricted to $m > 110$ GeV). Trigger turn-on modelling or the NLO Z pT spectrum; an in-situ trigger measurement is the next step.
4. The fake factor closes only with N_jets, era and $|\eta(\tau_1)| / pT(\tau_2)$ corrections; the FF is eta-dependent by +/-15 % because DeepTau's jet rejection is not. The MC subtraction is not optional (bias -0.05 on mu; unsubtracted fakes 7 % high where there is no signal).
5. 38 % of the selected Z->tautau is non-fiducial; kept in the signal template it makes the sideband mu-dependent. As a background it costs a 5 % normalisation NP and nothing else.
6. The BDT categories give a nearly background-free Z peak (S/B 5.1) and confine the fake uncertainties (9 %, now honest) to where the fakes are; the statistical gain is small (the measurement is systematics limited).
8. Tight working point (same chain): $\mu_Z = 1.071^{+0.114}_{-0.100}$, total +11/-10 % (Medium +15/-13 %), GoF $p = 0.25$: every uncertainty group shrinks, the visible-pT slope is weaker but still there. Recommended nominal for the next iteration and the combination.
7. Reproducibility: this checkout's TRExFitter was a v1.10 commit built against the StatAnalysis ROOT; it is now the tagged v1.8.0 built with fitting/build_trexfitter.sh.

For the combination

workspace fit/results/ztautau (job ztautau, POI mu_Z, regions tautau_SR0/1/2) plugs into fitting/combination_skeleton.config

correlated: Lumi, Pileup, L1Prefiring, QCDScale, PDF, PS_ISR/FSR, XS_*; tau-only: TauID/Trigger/ES_DM*, Fake*_tautau, XS_DYtautau_nonfid

SigModel_tautau is reported, not fitted; never correlate it with z-mumu's SigModel (a different quantity)

with ee/mumu fixing mu_Z, tautau constrains the tau ID nuisance parameters in situ (lepton universality): present TauID_DM* as a result

Born-level 60-120 GeV denominator for sigma; acceptance uncertainty 3.7 % correlated

Next steps

(CLAUDE.md, open issues)

decay-mode categories -> tau ID in situ even before the combination; quote the pT-binned POG prescription as a cross-check

trigger efficiency from mu tau_h tag-and-probe (SingleMuon) instead of the POG turn-on SFs

HT-binned W+jets; the high-mass DY sample for the non-fiducial template; EWK Z->tautau

add e tau_h / mu tau_h channels

Everything here is reproducible from the ntuples in ~25 min: python run_all.py --from 3 (z-tautau/README.md)